

Generation Interconnection System Impact Study Report

For Queue Project Q18-1039

Fair Oaks 138 kV
September 18, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1039. The Interconnection Customer proposes a 404.3 MW wind facility interconnecting at the Fair Oaks 138 kV substation in the Pennsylvania area. The request was received on August 16, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 262 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$62,466,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
FAIR OAKS – LYNBROOK 138 KV L	FAIR OAKS 345 kV/138 kV Autotransformer	242.1	200	100.0
LYNBROOK – HIGH ST 138 KV	FAIR OAKS 345-138 kV TX	236.0	204	100.0
COON RAPIDS II – HIGH STREET 138 KV Line	FAIR OAKS 345-138 kV Transformer	233.0	205	100.0
NOLENSVILLE – COON RAPIDS II 138 KV	FAIR OAKS 345-138 kV Autotransformer	197.4	240	100.0
NOLENSVILLE – SWEET 138 KV	FAIR OAKS 345/138kV Autotransformer	194.7	242	100.0
SWEET – COLLIERVILLE 138 KV L	FAIR OAKS 345-138 kV TX	189.0	247	100.0
DUNDEE – METRO CENTER 138 KV	CAMERON – ATHALMER TAP SUBSTATION 138 KV Line	139.0	333	5.2
FIRSTENERGY R PAUL SMITH POWER STATION – DAM NO. 4 HYDRO STA 138 KV	OTEGO – CARL BAILEY 138 KV	138.6	313	6.7
HOLLISTER – EL SOBRANTE 138 KV	PINEVILLE – E HAGERSTOWN 138 KV	136.4	411	5.8
INNERARITY – PINEVILLE 138 KV	PINEVILLE – E HAGERSTOWN 138 KV	131.4	458	5.8
BELLSFORK – MARYLAND SOLAR 138 KV	PINEVILLE – E HAGERSTOWN SUBSTATION 138 KV	131.3	449	5.8
MESSICK – PLUESS 138 KV	DUNDEE – METRO CENTER 138 KV line	131.1	438	6.9
CAMERON – PLUESS 138 KV	METRO CTR – DUNDEE 138 KV	130.4	455	6.9
ATHALMER TAP – CAMERON 138 KV line	DUNDEE – METRO CTR 138 KV	129.6	468	6.9

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
FIVE CHANNELS – MESSICK 138 KV	DUNDEE – METRO CTR 138 KV	127.3	430	6.9
ATHALMER TAP – E HAGERSTOWN 138 KV line	PINEVILLE – INNERARITY 138 KV	126.7	568	7.4
PINEVILLE – E HAGERSTOWN 138 KV line	INNERARITY – PINEVILLE 138 KV	124.4	605	7.4
EL SOBRANTE – BELLSFORK 138 KV line	PINEVILLE SUBSTATION – E HAGERSTOWN 138 KV	124.4	468	5.8
INNERARITY – MARYLAND SOLAR 138 KV	PINEVILLE – EAST HAGERSTOWN 138 KV	123.8	476	5.8
CHEVMAIN – CRESEY 138 KV L	WEST WAYNESBORO – CHAMBERSBURG 345 KV	121.1	645	7.0
CHEVMAIN – BUFFALO TUNGSTEN 138 KV	W WAYNESBORO – CHAMBERSBURG 345 KV	120.9	652	7.0
BUFFALO TUNGSTEN – CARREL 138 KV Line	W WAYNESBORO – CHAMBERSBURG 345 KV	120.6	660	7.0
FIRSTENERGY R PAUL SMITH POWER STATION SUBSTATION – CRESSEY 138 KV	CARL BAILEY – OTSEGO 138 KV	120.4	328	6.7
FAIR OAKS – CHAMBERSBURG 345 KV	CARREL 345/138kV XFMR	119.3	1,662	51.3
FAIR OAKS – CARREL SUBSTATION 345 KV Line	FAIR OAKS – CHAMBERSBURG 345 KV	116.6	1,524	77.8

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
FAIR OAKS – LYNBROOK 138 KV	Rebuild line with higher-capacity conductor	\$5,018,000	80% (\$3,998,000)
LYNBROOK – HIGH STREET 138 KV	Rebuild line with higher-capacity conductor	\$2,674,000	80% (\$2,130,000)
COON RAPIDS II – HIGH STREET 138 KV Line	Rebuild line with higher-capacity conductor	\$6,949,000	80% (\$5,537,000)
NOLENSVILLE – COON RAPIDS II 138 KV	Rebuild line with higher-capacity conductor	\$24,412,000	80% (\$19,451,000)
NOLENSVILLE – SWEET 138 KV	Rebuild line with higher-capacity conductor	\$8,668,000	80% (\$6,907,000)
COLLIERVILLE – SWEET 138 KV	Rebuild line with higher-capacity conductor	\$2,527,000	80% (\$2,013,000)
DUNDEE – METRO CTR 138 KV	Rebuild line with higher-capacity conductor	\$26,529,000	9% (\$2,503,000)
FIRSTENERGY R PAUL SMITH POWER STATION – DAM NO. 4 HYDRO STA 138 KV	Rebuild line with higher-capacity conductor	\$2,347,000	9% (\$216,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
HOLLISTER – EL SOBRANTE 138 KV	Rebuild line with higher-capacity conductor	\$4,585,000	9% (\$433,000)
PINEVILLE – INNERARITY 138 KV	Rebuild line with higher-capacity conductor	\$2,167,000	9% (\$205,000)
BELLSFORK – MARYLAND SOLAR 138 KV	Rebuild line with higher-capacity conductor	\$4,253,000	9% (\$401,000)
MESSICK – PLUESS 138 KV	Rebuild line with higher-capacity conductor	\$5,647,000	9% (\$533,000)
CAMERON – PLUESS 138 KV Line	Rebuild line with higher-capacity conductor	\$2,848,000	9% (\$269,000)
ATHALMER TAP – CAMERON 138 KV L	Rebuild line with higher-capacity conductor	\$4,075,000	9% (\$385,000)
FIVE CHANNELS – MESSICK 138 KV	Rebuild line with higher-capacity conductor	\$1,498,000	9% (\$141,000)
ATHALMER TAP – E HAGERSTOWN 138 KV	Rebuild line with higher-capacity conductor	\$4,886,000	9% (\$461,000)
PINEVILLE – E HAGERSTOWN 138 KV	Rebuild line with higher-capacity conductor	\$3,664,000	9% (\$346,000)
EL SOBRANTE – BELLSFORK 138 KV line	Rebuild line with higher-capacity conductor	\$5,491,000	9% (\$518,000)
INNERARITY – MARYLAND SOLAR 138 KV	Rebuild line with higher-capacity conductor	\$2,406,000	9% (\$227,000)
CHEVMAIN – CRESEY 138 KV L	Rebuild line with higher-capacity conductor	\$11,137,000	9% (\$1,018,000)
CHEVMAIN – BUFFALO TUNGSTEN 138 KV	Rebuild line with higher-capacity conductor	\$2,496,000	9% (\$228,000)
BUFFALO TUNGSTEN – CARREL 138 KV L	Rebuild line with higher-capacity conductor	\$1,539,000	9% (\$141,000)
FIRSTENERGY R PAUL SMITH POWER STATION SUBSTATION – CRESSEY 138 KV	Rebuild line with higher-capacity conductor	\$6,916,000	9% (\$633,000)
FAIR OAKS – CHAMBERSBURG 345 KV line	Rebuild line with higher-capacity conductor	\$54,652,000	14% (\$7,773,000)
FAIR OAKS – CARREL SUBSTATION 345 KV Line	Rebuild line with higher-capacity conductor	\$32,704,000	18% (\$5,999,000)
	Total allocated to project		\$62,466,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.